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Water-energy tradeoffs in data centers: A case study in hot-arid climates☆

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ABSTRACT

Although energy and water are intertwined, the analysis of energy consumption in data centers (DCs) has largely ignored water needs. In this study, we utilize established DC efficiency metrics, power usage effectiveness (PUE) and water usage effectiveness (WUE), to analyze the water and energy use in two colocation DCs (DC1 and DC2) located in the greater Phoenix area. These DCs use distinct cooling systems: DC1 uses air-cooled chillers and DC2 uses both water-cooled chillers and evaporative cooling. We also present a method for analyzing the water-energy tradeoff in DCs as a useful tool for DC designers and managers to evaluate different cooling systems. The results highlight that although DC1's average PUE is nearly 13% higher compared to DC2's average PUE, its source WUE is 66% lower than DC2. In addition, DC1's PUE is more affected by seasonal changes (variation of dry-bulb temperature) , while DC2's WUE has a larger seasonal variation. Our cooling system analysis indicates that hybrid evaporative cooling has the least power consumption, while air-cooled chillers have the most water use. Utilizing free cooling and evaporative cooling when outside air condition permits (~40% per year), results in substantial power and water savings in Phoenix DCs.

1. Introduction

The water-energy nexus plays a critical role in cloud infrastructure, however, the analysis of energy consumption in data centers (DCs) has largely ignored water needs. The total projected energy consumption in US DCs in 2020 was approximately 73 TWh (Shehabi et al., 2016) — which is the equivalent amount of energy used by nearly seven million residential homes. The COVID-19 pandemic has recently highlighted the role of the DC industry in worldwide economies, retail and supply chains, health care industry, educational systems, and personal and professional lives alike. Compared to pre-lockdown levels, internet services have increased 40–80%, and video conferencing with Zoom alone has increased nearly 250% (De et al., 2020, Koeze and Popper, 2020, Obringer et al., 2021). The trend in DC growth and energy consumption is only increasing (Jin et al., 2020, Koot and Wijnhoven, 2021, Obringer et al., 2021, Wei et al., 2021).

Energy consumption and efficiency in DCs has been the focus in their design and research. Nearly all of the electrical energy provided to information technology (IT) equipment is converted to heat, and must be effectively removed from the servers to prevent equipment failure (Zhang et al., 2017). The energy needed for powering servers and cooling IT equipment represent the largest costs in DC operation; DC cooling requirements alone represent upwards of 40% of the total energy use of DCs (ASHRAE, 2009). The type and design of cooling systems in DCs are influenced by several design criteria including local climate type, energy savings requirements, indoor environmental conditions, DC architecture, and capital and operational costs (Mytton, 2021). However, the dependence on water in these cooling systems and the important role of the water-energy nexus in DCs has been largely overlooked. For example, DCs consume approximately 9.1 liters of water per kWh of energy consumed (~20% on-site cooling and ~80% for grid electricity generation (Shehabi et al., 2016)) – amounting to nearly 660 billion liters of water used per year in US DCs.

Cooling requirements in DCs are further complicated by geographic location, climate type, and resource availability. The greater Phoenix area currently hosts approximatly 707 MW of DC IT capacity, and is expected to become the second largest DC market in the US, after Ashburn, North Virginia (Rubenoff, 2018). Both Phoenix and Ashburn are

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considered prime locations for DCs because of the effective power distribution, low energy rates, and positioning as they are both major fiber optic hubs connecting large cities. Although Phoenix boasts the additional benefit of having minimal threats from natural disasters, it is very water stressed, making energy efficient cooling that heavily rely on water unsustainable.

Cooling methods in DCs can be broadly categorized by three different technologies: (1) refrigeration cycles, (2) adiabatic cooling, and (3) free cooling. Although liquid-cooling is gaining interest, especially in enterprise DCs and high-performance computing centers with high rack densities, liquid cooling often requires additional infrastructure and specific IT equipment, thus limiting its application in colocation facilities. Cooling technologies can be used separately or in combination with others depending on the indoor and outdoor environmental conditions and have remarkably different energy and water consumptions.

Refrigeration cycles are common cooling technologies used in DCs; however, they are also very energy intensive. Two of the most common refrigeration cycles include direct expansion (DX) and chillers/vapor-compression. Furthermore, chillers/vapor compression systems can be categorized as air-cooled condensers (air-cooled chiller) or water-cooled condensers coupled with cooling towers (water-cooled chiller). Air-cooled chillers use outdoor air for heat rejection and are more energy-intensive (nearly zero water use), while water-cooled chillers use cooling tower water for heat rejection and are more water-intensive. Adiabatic cooling is another low-energy cooling technology; however, they can be water intensive technologies and are not applicable in hot-humid climates because of the high outside air (OA) wet-bulb temperature (Cui et al., 2014). Economization ("Free Cooling") systems include air-side economizers (ASE) and water-side economizers (WSE) and are generally coupled with another cooling system (i.e., refrigeration or adiabatic). ASE systems are advantageous when the outdoor air temperature is lower than the ASHREA recommended IT room temperature; whereas, WSE systems are advantageous when water leaving the cooling towers is available and cold enough to directly or indirectly reject heat from the IT room supply air.

While all cooling systems are designed to effectively provide desired IT conditions, there is often a trade-off between water and energy consumption. For example, hot-dry climates such as those in the arid Southwestern US are generally water stressed, and energy efficient cooling systems that consume large volumes of water for adiabatic cooling and/or cooling tower blowdown and make-up water may be impractical. Therefore, selecting the most appropriate DC cooling systems based on local climate constraints and resource availability is of great importance in moving towards more sustainable DCs.

1.1. Power and water usage effectiveness metrics

Because of the importance of power consumption in DCs, several organizations including government, non-profit agencies, manufacturers, and industry have worked to identify and promote energy efficient improvements, and have used power usage effectiveness (PUE) as a metric of DC energy efficiency performance (Avelar et al., 2012, Demetriou, 2012, Verdun, 2008). PUE, first proposed by the Green Grid Consortium, is defined as the ratio of the total facility energy to the IT energy (Belady et al., 2008):

$$\mathrm{PUE} = \frac{\text{Total Facility Energy}}{\text{IT equipment Energy}} \tag{1}$$

The total facility energy includes DC infrastructure systems and IT equipment energy. The energy needed for cooling the IT equipment represents one of the largest components of infrastructure systems energy. By definition, a PUE that approaches unity represents a complexly efficient DC with zero infrastructure system energy use. PUE is a commonly used metric to compare the performance of different DCs, enhance DC energy efficiency, determine the required energy for additional IT equipment, and meet design goals for new DCs (Avelar et al., 2012).

Water usage effectiveness (WUE) was introduced several years after PUE in 2011, also by the Green Grid Consortium, to address the water usage in DCs and aid in elevating DC sustainability (Patterson, 2011). WUE is defined as the ratio of onsite water usage (by volume) to IT equipment energy:

$$\mathrm{WUE}_{onsite}\left(\frac{\mathrm{L}}{\mathrm{kWh}}\right) = \frac{\text{Onsite Water Usage}}{\text{IT equipment Energy}} \tag{2}$$

The numerator in WUE accounts for the facility onsite water usage, which is primarily used in cooling and humidification systems. While the optimum value of PUE is unity, the optimum value for WUE is 0 L/kWh, meaning no water is used for DC operations. A more comprehensive WUE metric which accounts for DC onsite and source energy water usage (by volume) is $\mathrm{WUE}_{\mathrm{source}}$:

$$\mathrm{WUE}_{\mathrm{source}}\left(\frac{\mathrm{L}}{\mathrm{kWh}}\right) = \frac{\text{Onsite Water Usage} + \text{Source Water Use for Energy Generation}}{\text{IT equipment Energy}} \tag{3}$$

The annual source water use for energy production is defined as the annual amount of water used for producing the energy supplied to the DC.

$$\text{Source Water Energy Usage (L)} = \mathrm{EWIF} \times (\text{Total Facility Energy}) \tag{4}$$

The energy water intensity factor (EWIF)is defined as the amount of water used to produce energy. The estimated US nationwide average EWIF is 5.1 (L/kWh); however, EWIF can range from 0 to nearly 85 L/kWh based on geographic location and electricity generation process and operation (Peer et al., 2019; Siddik et al., 2020; 2021). For example, the Salt River Project (SRP), a major utility in Arizona, uses an average EWIF of 1.62 L/kWh, which is based on the electricity generation type breakdown in the SRP territory (Salt River Project power and water, 2021). Relative to the U.S. average, SRP's lower EWIF is attributed to several water conservation and cost/benefit savings measures implemented by SRP. For example, cooling tower water consumption is minimized by using increased cycles of concentration, and geothermal water losses are minimized by using flash steam generation. Also, the water loss for SRP's solar and hydroelectric energy generation is considered negligible. The large amount of water savings from cleaning solar panels outweighs the benefit of increasing panel efficiency. Additionally, evaporative water losses from hydroelectric reservoirs are attributed to the primary purpose of the reservoirs (i.e., residential, industrial, and agricultural water supply) rather than power generation.

1.2. Current study

The objective of this study is to analyze the water and energy use in two colocation DCs with distinct cooling systems located in the greater Phoenix area. We evaluate and compare the PUE, $\mathrm{WUE}_{\mathrm{onsite}}$, and $\mathrm{WUE}_{\mathrm{source}}$. We also analyze the water and energy trade-off and footprint for different DC cooling systems located in the greater Phoenix area, and

present a useful method for DC designers and managers to evaluate the water-energy trade-offs in cooling systems. Finally, we discuss the potential energy/water savings in DCs and how they impact the reliability of DCs. The results of this work provide (1) a holistic viewpoint of the water-energy nexus in DCs located in hot-dry climates, and (2) a new water-energy trade-off approach that DC planners and operators can use to identify more energy efficient modes of operation and make trade-off decisions between water and energy savings.

2. Methodology

The energy and water use for two colocation DCs (DC1 and DC2), located in Phoenix, AZ and served by SRP are presented. The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) climate type classification for this location is 2B: hot and dry. DC1 and DC2 have an overall IT capacity of 54 and 34.5 MW, respectively. In both DCs, the IT room condition is maintained in the ASHRAE recommended envelope, and the IT room cooling is performed by computer room air handler (CRAH) units that are located next to the data halls. The CRAH units reject heat generated in the IT room to the chilled water provided by the chillers. DC1 has multiple buildings with the same architecture and cooling system. In DC1, the CRAH units receive chilled water from a set of 500-ton air-cooled chillers that operate at their 75% nominal capacity. DC2 is contained in one building with a total of 20 suites or colocations. The entirety of DC2 is supported by a central water-cooled chiller plant consisting of multiple water-cooled chillers ranging from 400 to 800 tons in capacity (8200 tons total capacity) and four cooling tower plants (~7800 tons total capacity). However, in addition to the central water-cooled chiller plant, eight of the colocations utilize a hybrid cooling system consisting of direct evaporative cooling suites, or roof top units (RTUs). DC2's cooling system has the capability of using air-side economizer; however, at the time of the data acquisition, air-side economization was not being utilized.

2.1. Data acquisition & analysis

2.1.1. Power & water usage measurements in DC1 and DC2

2.1.1.1. Power usage in DC1. One-month of IT load data in October 2019 was provided by DC1. The dataset includes the IT equipment power use measured at the uninterrupted power supply (UPS) output. The IT load data provided is for a single building (Building A) consisting of two Data Halls with a nominal IT capacity of 18 MW. Thus, to compare the seasonal variations in energy and water efficiency of the two data centers the overall utility power use in Building A in DC1 was calculated using equipment manufacture data, meteorological data, and design building descriptors following ASHRAE's Standard 90.1–2010 Performance Rating Method (Pacific Northwest National Laboratory, 2016, Rasmussen, 2011). Detailed model inputs and calculations are summarized in the supporting information (Figure SI-1).

2.1.1.2. Power usage in DC2. IT equipment and utility power use at the UPS output and building entry, respectively, was provided for a 24-month period (January 2018-December 2019) for DC2. The data provided is representative of the entire DC. Among the 20 colocations in DC2, 12 colocations are connected to only the central chilled water plant, while the other eight colocations utilize both RTUs and the central chilled water plant.

2.1.1.3. Water usage in DC1. The air-cooled chillers circulate water in a closed loop; therefore, there is no additional cooling water usage in DC1. However, water is consumed for IT room humidification to meet ASHRAE's recommended humidity specifications. The onsite water usage measurements include the total water usage for all sources of water consumption at the entire site for six months from January to June of 2018. The individual building water use is assumed to be proportional to its IT capacity. Because of the small variation in onsite water use during the year, the overall monthly average water consumption is used in WUE calculations.

Table 1
Energy requirement for onsite water supply for both DCs. The surface water source supply is from the Colorado River and is part of the Central Arizona Project (Scott et al., 2011).

Parameter	Value	Ref.
Water supply to DCs (ground water: surface water)	50:50%	SRP
Surface water conveyance power intensity	1.2×10^6 kWh/m^3	(Scott et al., 2011)
Average ground water depth	100 m	(Arizona Municipal Water Users Association AMWUA, 1994, James and O'Dell, 2019)
Surface water treatment energy intensity	22×10^3 kW/m^3	(Water in the West, 2013)
Water distribution energy intensity	265×10^3 kW/m^3	(Water in the West, 2013)
Ground water pressurization level	4 bar (50 psi)	(City of Phoenix, 2021)
Pump efficiency	60%	(City of Phoenix, 2021)
Wastewater treatment energy intensity	330×10^3 kW/m^3	(Hamilton et al., 2009)

2.1.1.4. Water usage in DC2. Monthly water use for the entire DC over a 12-month period from January to December of 2019 was provided. In DC2, water is predominantly used for cooling towers, evaporative cooling, and humidification.

2.1.2. PUE and WUE in DC1 and DC2

The PUE (Eq. (1)) for both DCs was calculated at the basic level (Level 1) using the Green Grid Association's classification (Avelar et al., 2012). The basic level PUE accounts for the monthly IT equipment energy use measured at the UPS output (Figure SI-4).

Onsite WUE (WUE_{onsite}) and source WUE (WUE_{source}) were also calculated for both DCs. WUE_{onsite} (Eq. (2)) was calculated based on reported water usage in both DCs. WUE_{source} was calculated using an EWIF of 1.62 L/kWh as indicated by SRP.

2.1.3. PUE sensitivity analysis for DC1

Because air-cooled chiller energy efficiency is highly sensitive to outside air dry-bulb temperatures and the chillers' power use significantly affects the PUE, sensitivity analyses were performed for DC1. Three monthly average dry-bulb temperatures (minimum, average, and maximum) for Phoenix, AZ from January 2019 to December 2019 were used to evaluate DC1's chiller operating performance and PUE at moderate and maximum weather conditions (The Weather Company, 2021). Air-cooled chillers usually run between 22% and 100% of their nominal capacity (Alonso et al., 2019), where DC1's chillers operate at a minimum of 75% nominal capacity. Thus, the chiller performance capacity was varied from 70 to 100% percent in the sensitivity analysis.

2.2. Enhanced water and energy utilization effectiveness metrics

Although PUE can be a useful metric to compare energy efficiency in DCs, it does not include the intrinsic power used for water conveyance and treatment. Therefore, we propose the enhanced PUE (PUE_e) as a comprehensive metric that accounts for the required energy use for onsite water usage in addition to the standard PUE.

$$PUE_e = PUE + \frac{Required\ Energy\ for\ Onsite\ Water\ Usage}{IT\ equipment\ Energy} \quad (5)$$

Similar to PUE, the optimum value of PUE_e is unity. The required energy for onsite water use considers the energy required for source

Table 2
Average annual household water and energy use and utility scale energy and water costs for the greater Phoenix area.

Approximation	Value	Reference
Average household energy use (kWh/household)	11,564	(Southwest Energy Efficiency Project SWEEP, 2016)
Average household water use (m^3/household) *	444	(Arizona Department of Water Resources, 2021, United States Census Bureau, 2021)
Energy cost ($/kWh)	0.1	(Citty of Chandler Arizona, 2021)
Water cost ($/$m^3$)	0.55	(Citty of Chandler Arizona, 2021)

* Calculated based on average annual water use per capita per day of 454 L (120 gallons) and average household size of 2.68 persons (Arizona Department of Water Resources, 2021, United States Census Bureau, 2021).

water treatment and conveyance to the site. Data used to calculate the required energy for onsite water supply for both DCs is provided in Table 1.

While WUE_{source} includes the intrinsic water use for energy generation, it is not a dimensionless metric that can be easily compared. Therefore, we propose enhanced WUE (WUE_e) as a dimensionless, more comprehensive metric with the optimal value of unity to compare the water-energy trade-off in DCs at both the localized and regional scale.

$$WUE_e = \frac{\text{Onsite Water Usage} + \text{Source Water Use for Energy Generation}}{IT\ equipment\ Energy \times EWIF} \tag{6a}$$

$$WUE_e = \frac{WUE_{source}}{EWIF} \tag{6b}$$

Because EWIF is in the denominator, the local and regional water and energy resources must be considered in the analysis. For example, this metric can be beneficial in comparing the water efficiency of different DC configurations in localized regions with similar power sources and EWIF values. This metric can also be beneficial in comparing the regional water efficiency of DCs with the same configuration, as in the case with larger enterprise and colocation DC providers, in different geographic regions. Because DC1 and DC2 are in the same geographic region with the same EWIF, the water efficiency of different DC configurations can be compared.

2.3. Water and energy usage and trade-off in phoenix DCs

To gain a better understanding of the impacts that DCs have on local water and energy resources, a hypothetical scenario estimating the total water and energy usage for Phoenix's existing IT capacity of 707 MW using solely one of four different cooling technologies was evaluated. The four cooling technologies include: (1) an air-cooled chiller, (2) a water-cooled chiller, (3) a hybrid air-cooled chiller with direct evaporative cooling, and (4) a hybrid water-cooled chiller with direct evaporative cooling. The main differences between the cooling technologies are the use of either air or water cooling for the chiller with or without evaporative cooling. The energy and water consumption for the four cooling processes were adopted from a 2013 study referencing modeled cooling systems from United Metal Products, a leading manufacturer in HVAC systems (Vokoun, 2021). In this analysis, the outside air supplied to the IT room (36,000 cfm) is cooled to 26.7 °C to meet ASHRAE's recommended temperature range (Table SI-3). The available hours of free cooling is calculated using hourly meteorological data from the EnergyPlus TMY3 weather database for Phoenix-Sky Harbor International Airport 722,780 (Wilcox and Marion,). In 2019, the outside air temperature was below 26.7 °C for approximately 5450 h. Additionally, an EWIF of 1.62 L/kWh was used in this analysis. The variations in predicted water and energy usage for the four scenarios was calculated by referencing the power breakdown in five DCs with different IT capacities (Demetriou, 2012, Mahdavi, 2013, Nair, 2011, Vokoun, 2021). The estimated displacement on local water and energy resources was evaluated by calculating the total annual energy (GWh) and water (million m^3) consumed by the four hypothetical scenarios based on data from five case studies and DC1 and DC2. Eqs. (5) and (6b) were used to calculate the PUE_e and WUE_e and evaluate the water- energy trade-off. The source water and source power were treated as the same in this analysis, and the assumptions listed in the supplementary information apply. The total costs and equivalent number of households served were calculated using the approximations in Table 2. The approximations are for the greater Phoenix area, and the referenced costs are commercial rates.

2.4. Energy and water conservation analysis

A sensitivity analysis was performed on expanding ASHRAE's recommended operating enveloped (Figure SI-6) (ASHRAE Technical Committee 9.9, 2012) to increase energy and water savings. The change

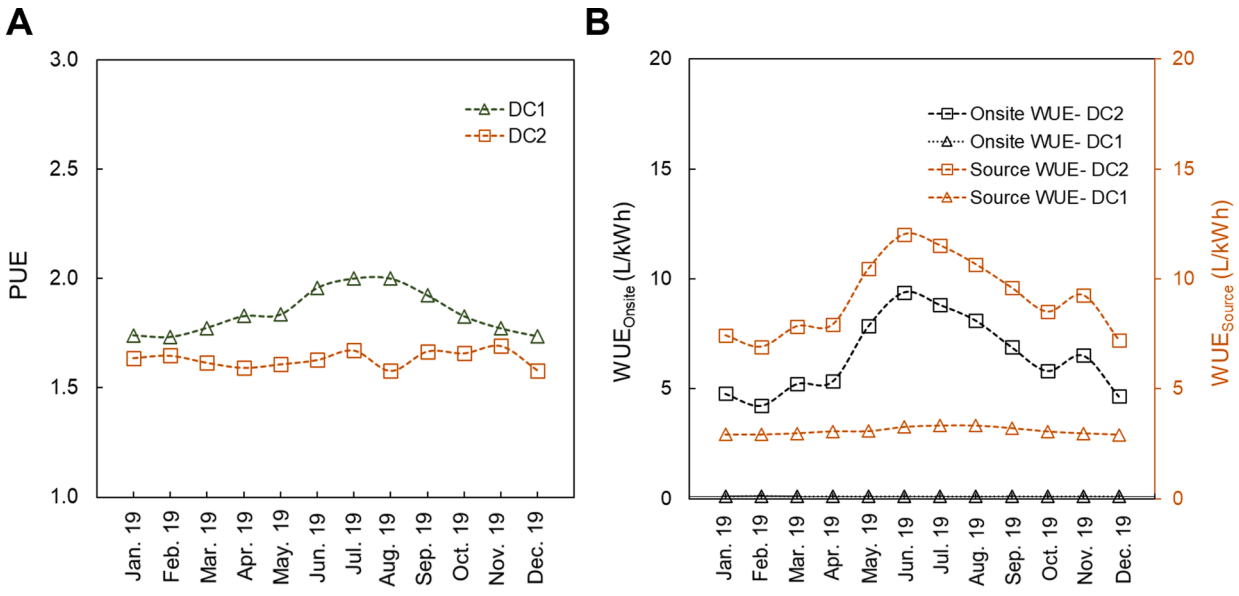


Fig. 1. Temporal variations of (A) basic PUE and (B) onsite and source WUEs for DC1 and DC2.

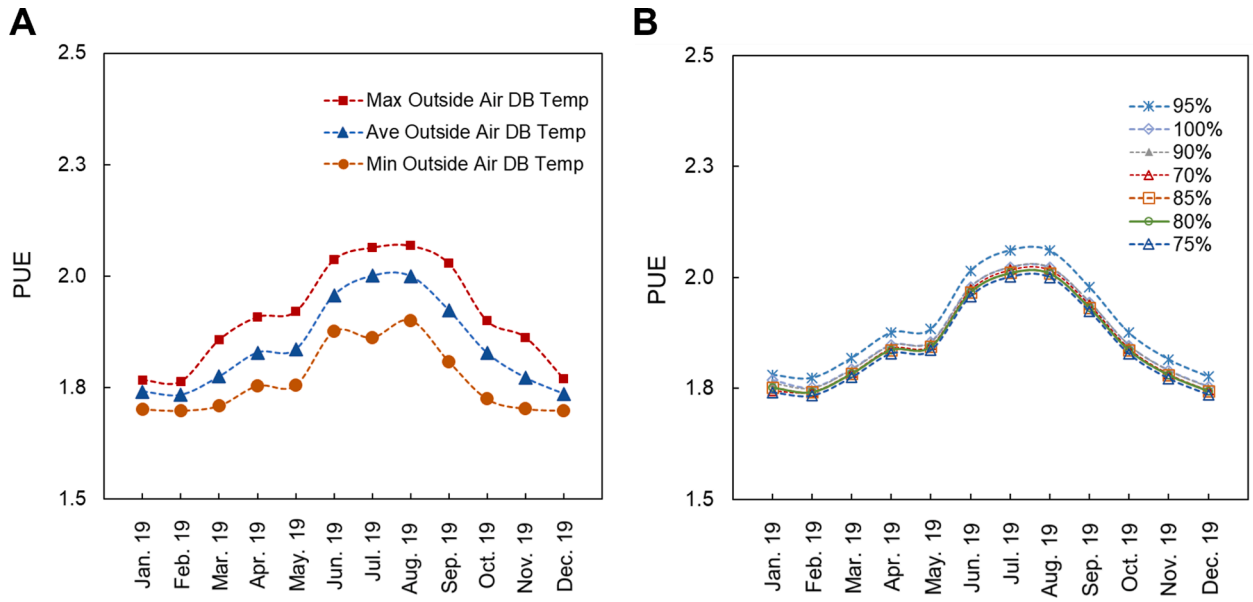


Fig. 2. Effects of (A) outside dry-bulb temperature and (B) chiller and CRAH loading factor (capacity) on DC1 PUE.

in failure percentage when the IT room envelopes are changed can be evaluated and predicted using a useful parameter defined by ASHRAE known as the reliability X-factor (ASHRAE Technical Committee TC 9.9 Mission Critical Facilities, Data Centers, Technology Spaces, and Electronic Equipment, 2016). For example, a reliability X-factor of 1.3 (recommended IT room envelope with a dry-bulb temperature of 27 °C) means that there is a 30% higher probability of server failure compared to operating the DC at 20 °C (server failure rates of 0.2%).

3. Results and discussion

3.1. PUE & WUE

Monthly variation of basic PUE as well as onsite and source WUE for DC1 and DC2 are shown in Fig. 1. DC1's PUE is consistently higher than DC2's PUE. DC1's PUE ranges from 1.73 to 2.00, has an average of 1.84, and peaks in the summer months; in contrast, DC2's PUE ranges from 1.58 to 1.69, has an average of 1.63, and is more consistent throughout the year because of the hybrid cooling system and lower variation in outdoor air wet-bulb temperature, which affects the water-cooled chiller power use. The higher PUE in DC1 is because it uses an air-cooled chiller which is more energy intensive than water-cooled chillers and hybrid cooling systems (Fig. 1A). In addition, the higher and larger variation in DC1's PUE is because the refrigerant condensing temperature in air-cooled chillers depends on dry-bulb temperatures which are considerably higher than wet-bulb temperatures and are more varied throughout the day and year. In water-cooled chillers, the refrigerant condensing temperature depends on the condenser water (cooling tower water) temperature which is influenced by the wet-bulb temperature. Based on Phoenix meteorological data during 2017–2019, the average and standard deviation of monthly dry-bulb temperature are 24.9 °C and 8.3 °C, respectively, while the average and standard deviation of monthly wet-bulb temperatures are 14.1 and 5.1 °C, respectively. Because the wet-bulb temperature is nearly 10 °C lower than the dry-bulb temperature,

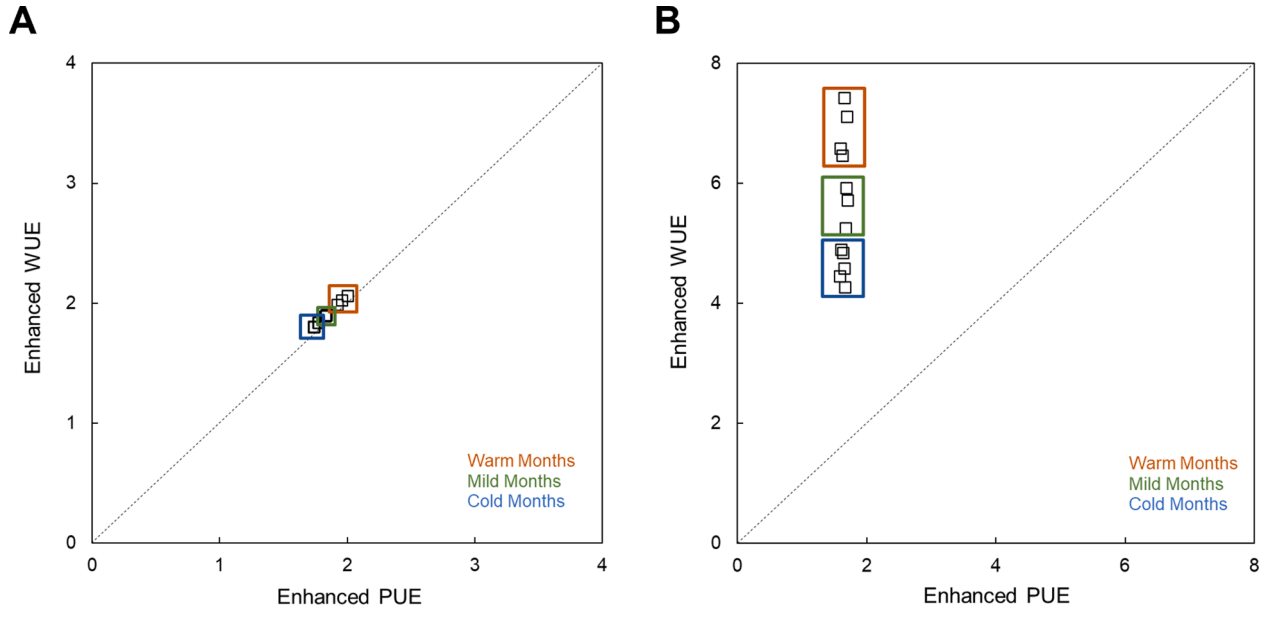


Fig. 3. Water-energy tradeoff in (A) DC1 and (B) DC2.

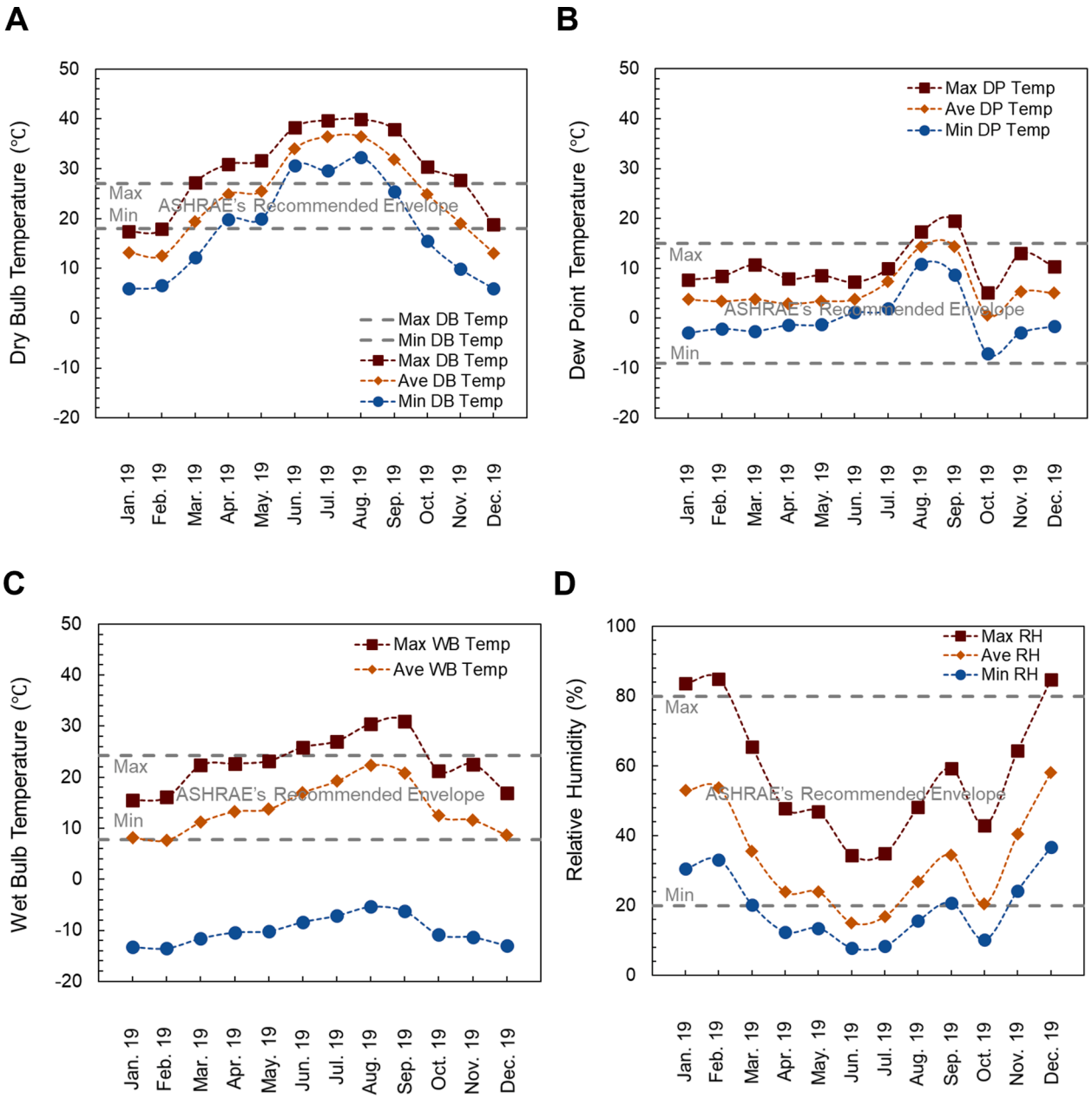


Fig. 4. Air-side and water-side economizer utilization potentials in Phoenix DCs; (A) Dry-bulb temperature monthly average, (B) Dew point temperature monthly average, (C) Wet-bulb temperature average, and (D) Relative humidity monthly average.

there is more opportunity for hours of water-side economization; however, this comes at the expense of increasing water consumption. Additionally, the dry-bulb temperature depression at night is larger than the wet-bulb compression, partially compensating for the lower energy efficiency of air-cooled chillers during the day (Wieman, 2019).

Although the PUE for DC2 is lower than DC1, by comparison, both the WUE_{Onsite} and WUE_{Source} for DC2 is larger because of the use of a water-based cooling system (Fig. 1B). DC2's WUE_{Source} has an average of 9.11 L/kWh and ranges from 6.92 to 12.03 L/kWh, peaking in June due to the increased evaporation rate of open loop cooling tower water, when the average dry-bulb temperature and minimum relative humidity were 34°C and 15.1%, respectively. Additionally, DC2's WUE_{Onsite} and WUE_{Source} have larger seasonal variations, and nearly 71% of the water is used onsite. In comparison, DC1's WUE_{Source} has an average of 3.10 L/kWh and ranges from 2.91 to 3.34 L/kWh. Despite DC1's relatively constant WUE_{Onsite}, WUE_{Source} demonstrates an increasing trend with the approaching hot season, mirroring its trend in PUE. Nearly 96% of DC1's water use is from source water for energy generation, where the remaining 4% of onsite water is used for humidification and landscaping. Although DC2's WUE_{Onsite} and WUE_{Source} are larger than those of DC1, the average source water energy use in DC1 (2.99 L/kWh) is higher than in DC2 (2.64 L/kWh). The larger source water energy use in DC1 is because of the energy intensive air-cooled chiller system. This analysis of energy and onsite/source water use in DC1 and DC2 directly highlights the water-energy trade-off in DC cooling system infrastructure.

3.2. Sensitivity analysis of PUE

The monthly variation in PUE for different outside dry-bulb temperatures (average-minimum, average-maximum, and average-average) and percent chiller and CRAH loading factor are shown in Fig. 2. For each sensitivity, PUE is highest during the hot summer months (June through August). In general, increasing the chiller and CRAH loading

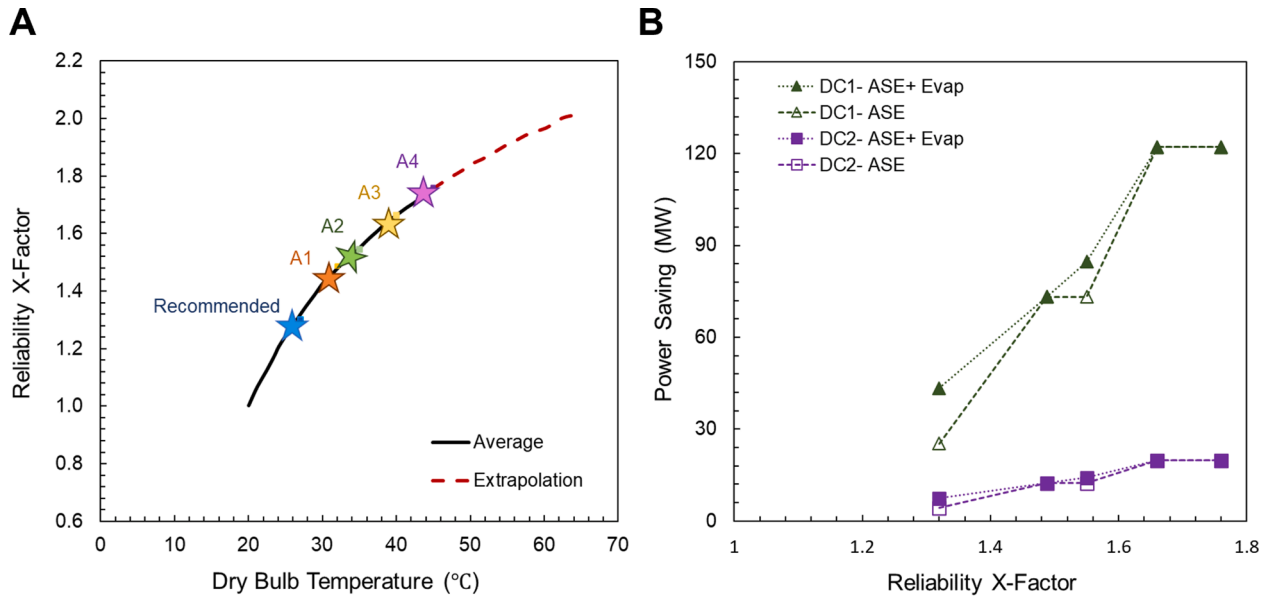


Fig. 5. Reliability X-factor as a function of dry-bulb temperature (adopted from (ASHRAE 2012), (A), and power saving using air-side economizer in two case studies DCs as a function of reliability X-factor (B).

factor has a marginal effect on PUE, and the small variation in PUE can be attributed to the variation in chiller coefficient of performance (Figure SI-3). Most of the chillers have a peak operating efficiency at 70–75% of their nominal loading factor (Clark, 2005); thus, increasing the chiller and CRAH loading factor above 75% generally results in increased PUE (Fig. 2B).The maximum standard deviation among DC1's PUE values at the three temperature levels and chiller loading factor is 0.11 and 0.02, respectively. Therefore, the outside air dry-bulb temperature has a larger impact on PUE than chiller and CRAH loading factor. A similar sensitivity analysis was performed for WUE_{source} (Figure SI-5). Although the onsite water usage does not change, as expected, the same trends were observed because the source water energy use is directly proportional to the total facility energy use.

3.3. Water-Energy trade-off in DC1 and DC2

WUE_e as a function of PUE_e for DC1 and DC2 are shown in Fig. 3. The dashed line on the graphs represents the water-energy tradeoff: a value above the diagonal line indicates the DC is more water intensive, a value below the line indicates the DC is more energy intensive, and a value on the line indicates the DC water and energy are directly proportional. The temporal impacts on PUE_e and WUE_e are further displayed when values are grouped by seasonal conditions (cool, mild, and hot months). For instance, WUE_e and PUE_e both have the highest values during hot seasons, while their values decrease in cold seasons. The average WUE_e and PUE_e for DC1 are 1.91 and 1.84 respectively. The average WUE_e and PUE_e for DC2 are 5.63 and 1.64 , respectively.

WUE_e is directly proportional to PUE_e in DC1; as the air-cooled chiller power consumption increases in warmer months to meet the required IT equipment cooling load, the source water usage and thus WUE_e directly increases. Depending on the time of year, the PUE_e for DC1 is nearly 12% larger than that of DC2; however, the WUE_e for DC2 is two to three times higher than that of DC1 during the same season. Similar to DC1, the WUE_e for DC2 increases as the temperature increases due to increased water use in the cooling towers. However, there is no seasonal variation in PUE_e for DC2. Although it was not possible to dissociate the power and water consumptions by different cooling systems because of lack of metering locations, we believe that during hot months, the onsite water consumptions from the cooling towers and adiabatic evaporative suites increase, which result in higher values of WUE_e. On the contrary, in cold seasons, the onsite water usage decreases.

3.4. Energy and water conservation approaches in DCs

An opportunity for saving energy and water in existing and future DCs in Phoenix is utilizing air-side economizer and adiabatic evaporative cooling. A system that uses air-economization for cooling is dependent on the dry bulb temperature and relative humidity, where a system that uses water-side economization for cooling is dependent on the wet-bulb temperature. ASHRAE's recommended IT room dry-bulb and dew point temperature range from 18 °C to 27 °C and −9 °C to 15 °C, respectively (Fig. 4A and 4B). Phoenix's monthly maximum dry-bulb temperature and the monthly mean ambient dry-bulb temperature are less than or equal to 27 °C for nearly 25% and 67% of the year, respectively (Fig. 4A). The monthly maximum dew point temperature and the monthly mean dew point temperature are in the recommended range for nearly the entire year (Fig. 4B). Therefore, there is great opportunity for relying entirely on air-side economizers for DC cooling for at least one-fourth of a year (Fig. 4A). In addition, if the ambient dry-bulb temperature is lower than ASHRAE's minimum recommended IT room dry-bulb temperature, mixing IT room return air with ambient air can further improve air filtering energy and humidification water savings. Furthermore, the use of an air-side economizer could be extended if used during specific hours of the day (i.e., during nighttime hours in the summer) or when hybridized with evaporative cooling effects. However, it is important to note that frequently changing the cooling mode can be operationally challenging and affect the equipment lifetime.

Based on Phoenix's monthly wet-bulb temperature and relative humidity (Fig. 4C-D), the maximum wet bulb temperature and relative humidity are less than the IT room maximum temperature (27 °C) and relative humidity (80%), respectively. Therefore, there is high potential of utilizing the adiabatic evaporative cooling system for nearly two-thirds of the year. Therefore, in addition to air-side economization, there is even more opportunity to lower energy and water consumptions using adiabatic evaporative cooling systems. However, use of adiabatic evaporative cooling systems over or in addition to air-cooled systems comes at the expense of increased water consumption. Additionally, if water-cooled chillers are used, water-side economizers can be partially utilized during the year under certain conditions (i.e., when the cooling

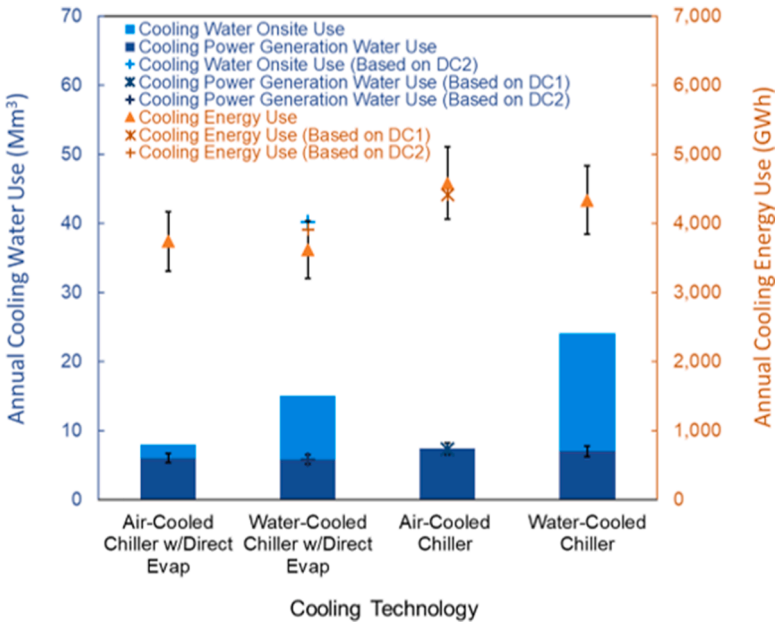


Fig. 6. Water and energy use for various cooling technologies supporting the existing 707 MW of IT capacity in Phoenix DCs.

tower water temperature approaches the chilled water setpoint). Given the abundant potential for renewable energy (i.e., solar) and stress on local water resources, making use of air-side and water-side economization and adiabatic evaporative cooling system could strike a good balance between energy and water savings in Phoenix-based DCs.

Additional DC water and energy savings are obtainable if DCs IT room operating envelopes are extended from A1 to A4. Increasing dry-bulb temperatures, results in increased energy savings for both DC1 and DC2 using air-side economizers at the cost of higher failure rate (Fig. 5A-B). In general, the trend of energy saving in DC1 by increasing the free cooling percentage in this DC is higher than for DC2 because of its higher PUE and high cooling energy usage. Operating at the ASHREA recommended IT room temperature of 27 °C results in the potential use of air-side economization for at 25% of the year and annual energy savings of 219 and 39.5 GWh in DC1 and DC2, respectively. The addition of adiabatic evaporative cooling results in an increased annual energy savings of about 70% in both DC1 and DC2. Furthermore, operating at the expanded A4 IT room envelope (i.e., dry-bulb temperature of 47 °C) results in an annual energy savings of 122 and 20 MW in DC1 and DC2, respectively. However, expanding the IT room operating envelope increases the failure rate to nearly 80%.

3.5. Water and energy usage and tradeoff in phoenix DCs

The type of cooling technology selected for DCs has major implications for onsite and off-site water and energy use, especially in water-stressed regions such as Phoenix, AZ. To gain a better understanding of the impacts that DCs have on local water and energy resources, the total DC water and energy usage for Phoenix's existing IT capacity of 707 MW using solely one of four different cooling technologies was evaluated (Fig. 6). The main difference between the cooling technologies is the use of air- and water-cooled chillers hybridized with or without evaporative cooling. The main trends for each of the cooling systems are summarized:

1) Air-cooled chillers with direct evaporative cooling consume the 2nd least amount of onsite water, and most of the water is consumed at the source (i.e., power plant). This cooling technology consumes the second to least amount of energy and is only slightly more energy intensive than the water-cooled chiller with direct evaporative cooling. Both direct evaporative cooling technologies have lower energy consumption compared to chiller technologies because they use the latent heat of vaporization of water to achieve cooling.
2) Water-cooled chillers with direct evaporative cooling consume the 2nd largest amount of water, and most of the water is consumed onsite at the cooling towers. Because of the use of evaporative cooling, this technology uses the least amount of energy.
3) Air-cooled chillers are the most water efficient and do not consume onsite water, however, they consume the most source water energy use. Air-cooled chillers consume the most energy because of the refrigerant's condensing temperature dependency on the dry-bulb temperature, which is usually considerably higher than the wet-bulb temperature.

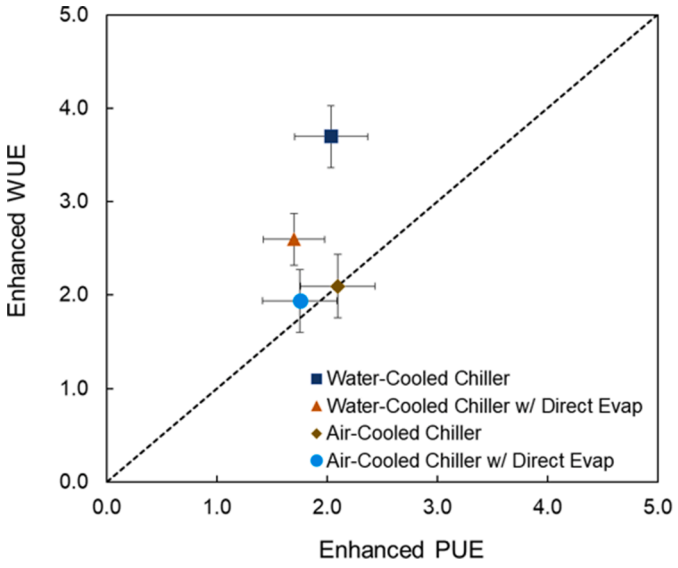


Fig. 7. Water-energy tradeoff for different cooling processes supporting the existing 707 MW of IT capacity in Phoenix DCs. The vertical and horizontal error bars represent the variation in overall WUE_e and PUE_e which were calculated based on the different power breakdown structures in DC1 and DC2 and five DCs with different IT capacities.

Table 3
The total annual cooling energy and water consumed by different cooling processes supporting the existing 707 MW of IT capacity in Phoenix DCs.

		Air-cooled chiller		Water cooled chiller	
		Direct evaporative cooling	Without evaporative	Direct evaporative cooling	Without evaporative
Energy	Total annual cooling energy use (GWh)	3,740	4,585	3,620	4,335
	Cost ($ million)	374	459	362	433
	Households served annually (% households*)	323,400 (53)	396,500 (65)	313,000 (52)	374,800 (62)
Water	Annual water use for onsite/source cooling ($m^3 \times 10^6$)	2/6	0/7.5	9/6	17/7
	Cost ($ million)	4.4	4.1	8.2	13
	Households served annually (% households*)	17,900 (3)	16,700 (2.7)	33,500 (5.5)	53,400 (8.8)

* The percent households served is based on the population of 1.633 million in Phoenix in 2019 (United States Census Bureau, 2009) and an average household size in Arizona of 2.68 persons (United States Census Bureau, 2021).

4) Water-cooled chillers consume the most onsite water, and thus total water for cooling, and have the 2nd highest energy consumption. Compared to the air-cooled chillers, using water instead of air results in small additional energy savings.

The water-energy tradeoff chart (Fig. 7) further indicates that air-cooled chillers with direct evaporative cooling result in both water and energy savings. Although the energy consumption for air-cooled chillers is higher than water cooled chillers, the PUE_e of air-cooled chillers with direct evaporative cooling is only slightly higher than the PUE_e of water-cooled chillers with direct evaporative cooling because of the high efficiency of evaporative cooling in hot-dry climate. However, the PUE_e of air-cooled chillers with direct evaporative cooling is not significantly different from the PUE_e of water-cooled chillers with direct evaporative cooling. The onsite water use in water-cooled chillers is the highest; more energy is required for water conveyance, treatment, and waste management which results in a larger value in the second term of PUE_e (Eq. (5)). The tradeoff between reduced energy consumption and increased water consumption is not directly proportional for all technologies. For instance, when compared to air-cooled chillers, water-cooled chillers offer a 3% reduction in PUE_e, however, they result in a 76% increase in WUE_e. Based on the water and energy tradeoff analysis, air-cooled chillers with direct-evaporative cooling result in the lowest PUE_e and WUE_e. The large discrepancy in WUE_e between the case studies and modeled data is likely because of DC inefficiencies including air flow management systems, dated cooling systems, IT equipment inefficiencies, airtightness, administrative water use, etc. Therefore, additional modeling efforts on cooling system water usage are needed.

The local water and energy resource displacement for the different cooling technologies supporting the 707 MW of IT capacity in Phoenix DCs is summarized in Table 3. It is important to note that the resource displacement is based solely on the energy needed for cooling the IT equipment and does not include the IT energy and other facility energy use. Water-cooled chillers without direct evaporative cooling have the second highest energy use and largest total water use; the annual energy and water use are equivalent to displacing approximatly 9% and 62% of Phoenix households from water and energy, respectively. Comparatively, air-cooled chillers with direct evaporative cooling narrowly have the second lowest energy and water use; the annual energy and water use is equivalent to displacing 3% and 53% of of Phoenix households from water and energy, respectively.

4. Conclusion

In this work, we analyzed the power and water use by two colocation DCs, using two different cooling systems, in the greater Phoenix area. The conclusions on power and water efficiency in Phoenix-based DCs are as follows:

- The air-cooled chiller in DC1 is more energy intensive than the water-based cooling system in DC2; however, DC1s water footprint is nearly one-quarter of DC2.
- DC1's PUE is more affected by seasonal changes (variation of dry-bulb temperature) compared to DC2's PUE, while DC2's WUE has a larger seasonal variation.
- In the case of DC1, the temporal impacts on PUE_e and WUE_e are similar because WUE_e is mostly affected by PUE and the onsite water use in DC is not significant.
- During hot months, the WUE_e for DC2 increases as the dry-bulb temperature increases due to increased onsite water consumption from the cooling towers and adiabatic evaporative suites. However, during cold seasons, the WUE_e for DC2 decreases because of reduced onsite water usage.
- There is a great opportunity for utilizing air-side economizers for cooling DCs in the greater Phoenix area. In addition, if the ambient dry-bulb temperature is lower than ASHRAE's minimum recommended IT room dry-bulb temperature, mixing IT room return air with ambient air can further improve energy and water savings. However, operating outside of ASHRAE's recommended envelopes, while offering additional energy savings, results in increased IT equipment failure.
- There is also great opportunity for using waterside economization in water-based cooling systems due to the lower wet-bulb temperature on an annual basis.
- Among the four distinct cooling systems analyzed to hypothetically support the existing 707 MW of IT capacity in Phoenix DCs, air-cooled chillers with direct evaporative cooling have the lowest combined energy and water footprint. While water-cooled chillers without direct evaporative cooling have the highest combined energy and water footprint displacing the equivalent water and energy supply to Phoenix households' by approximately 9% and 62%, respectively.

CRediT authorship contribution statement

Leila Karimi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Leeann Yacuel:** Resources, Supervision, Writing – review & editing. **Joseph Degraft- Johnson:** Resources, Writing – review & editing. **Jamie Ashby:** Resources, Writing – review & editing. **Michael Green:** Resources, Writing – review & editing. **Matt Renner:** Data curation, Validation, Writing – review & editing. **Aryn Bergman:** Data curation, Validation, Writing – review & editing. **Robert Norwood:** Conceptualization, Funding acquisition, Methodology, Resources, Supervision, Writing – review & editing. **Kerri L. Hickenbottom:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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